Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 165 (2019) 708–716

Procedia Computer Science

www.elsevier.com/locate/procedia

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019, ICRTAC 2019

KBR: Knowledge Based Reduction Method for Virtual Machine Migration in Cloud Computing

Bhaskar R*, Shylaja B S

Dr. Ambedkar Institute of Technology, Bengaluru, India.

Abstract

Through large scale deployment of cloud data centers, burning up of energy by data center is very high and violations of service level agreements (SLA) are increasing. Implementation of Virtual machine consolidation techniques direct to physical host over exploitation which degrades the performance of physical host and Virtual machine. It has been a momentous confronts to get better exploitation of resources and energy effectiveness in cloud data center. This paper proposed knowledge based reduction method for Virtual machine assignment and consolidation. This method chooses the Virtual machines from the overloaded host and migrate the Virtual machines to other physical host to diminish energy burning up and get better quality of service. To provide evidence and effectiveness of proposed method, the experiment is conducted with diverse workload circumstances of real system. The experimental consequences exhibits that the proposed method diminishes energy burning up and optimizes infringement of service level agreements in achieving better performance.

Keywords: Cloud Computing, Cloud-sim, Data center, Physical machine, Rough-set, Virtual machine

1. Introduction

Data centers supports large scale of Internet services and the operation cost of these data centers increases due to burning up of high energy. Physical resources of data centers e.g., CPU, Memory, Disks energy burning up and

*Corresponding author. Tel.: 91-8123422306
E-mail address: bhaskar@dbit.co.in

10.1016/j.procs.2020.01.026

service expenses are increased in the last few years. At present data centers use virtualization technology widely in most Physical machines. Whenever user requests resources are bundled as Virtual machines and positioned in diverse Physical machines (Hosts) based on service provider and users service level agreements, it improves resource exploitation. Each Virtual machine is bundled with certain quantity of resources e.g., CPU, Memory, Disk and Bandwidth[19]. Virtualization technology supports application performance, and numerous Virtual machines (VMs) can run on the same physical host, this help in improving resource exploitation and diminish energy burning up in data center. Additionally, virtualization technology can also facilitate cloud providers in order to set up resources on demand, offer an efficient elucidation to the elastic resource administration and diminishes energy burning up. On the other hand, unnecessary VM migrations bring in additional administration cost, e.g., creation and reconfiguration of Virtual machines, destruction of Virtual machines causes superfluous energy burning up. As a result, this work endeavors to diminish the numeral of Virtual machine migrations to diminish energy burning up.

To diminish energy burning up, this paper proposed rough-set based reduction method for Virtual machine assignment and consolidation method. In terms of energy burning up, proposed method enhances the performance compared to existing policies, migration time of Virtual machine, and violation of service level agreement percentage. The initial circumstance for Virtual machine migration is, the resource exploitation exceeds a convinced value, and Virtual machines on the Physical machine cannot convene service level agreement among consumers and service providers. This paper concentrates on setting of superior approximation of CPU exploitation to keep away from overloaded physical hosts and sustain the service level agreement. Later, Physical machine with low CPU exploitation are turned to standby mode, if Physical machines resource exploitation is lower than the inferior approximation, the Virtual machines on that physical host are migrated. Thereafter idle Physical machine is turned to standby, ensuing in smaller numeral of active hosts of which each one resource is well exploited. This work concentrates both of above scenarios on active consolidation of Virtual machines in which Virtual machines are every so often reallocated.

2. Related Work

Researchers approached the problem of Virtual machine consolidation and migration in Cloud, various resource allocation methods have been proposed for cloud computing to provide cost effective service.

Hui Xiao and et.al., proposed multi objective Virtual machine consolidation method [1] by using double threshold and ant colony system. When the resource rivalry turns into more concentrated specifically resource exploitation exceeds a convinced thresholds, performance of the Physical machine turn worse, and QoS is declined. The double threshold method is draw on to make judgment of the consolidation circumstances that trigger Virtual machine consolidation and optimized in the course of a multi-stage based on ant colony system.

Xijia Zhou and et.al., proposed empirical forecast algorithm and weighted Virtual machine selection algorithm [2], when resource exploitation exceeds the total capacity allocates to that Physical machine, algorithm predicts historical exploitation of the host and weight several exploitation reasons and compute its migration priority of each overloaded host. The Virtual machine manager performs the Virtual machine migration.

Hui Wang and et. al., proposed new framework for migration based on space aware best fit decreasing algorithm and high CPU exploitation based migration technique called DVMC (dynamic Virtual machine consolidation) [3]. This framework consists of four stages, monitoring, workload analysis, decision and actuation. A dynamic Virtual machine consolidation plan contains pronouncement on the placement of Virtual machines on physical hosts and the Virtual machine migration policy.

Yingyou Wen and et.al., proposed virtual resource dynamic integration (VRDI) method for live VM migration [4]. This method uses lower threshold and upper threshold to trigger the Virtual machine migration. When CPUs exploitation exceeds upper threshold or exploitation is lesser than lower threshold, triggers VM migration using live migration technology, which reduces the energy burning up of a data center by integrating the virtual resources.

Nidhi Jain Kansal and et.al., proposed Energy aware Virtual machine migration for cloud computing using firefly algorithm [5] . This technique migrates overloaded Virtual machines to a least loaded active host. This algorithm employs on artificial bee colony based energy conscious resource exploitation technique to offer the essential resources to the consumers applications to get better resource exploitation and to diminish the energy burning up in the cloud computing environment.

Nguyen Khac Chien and et.al., proposed live migration technique for Virtual machine migration in cloud computing [6]. When the resource rivalry turns into more concentrated specifically resource exploitation exceeds a convinced thresholds, this technique choosing VM that entails minimum time for migration and the left over available CPU is minimal to trigger the Virtual machine migration, leads to less occurrence of SLA violation.

Xiong Fu and et.al., proposed an algorithm called layered Virtual machine migration [7]. This algorithm uses minimum spanning tree theory, divides the data center into a numeral of regions based on the network bandwidth exploitation rate of the physical hosts. Then each region network resources are balanced by Virtual machine migration in the data center and achieve load balancing.

Abdelkhalik Mosa and et. al., proposed genetic algorithm to optimize Virtual machine assignment in cloud computing environment [8]. This method provides a solution for VM assignment in cloud data centers that vigorously assigns Virtual machines to hosts based on the resource exploitation. This method increases the profit and diminishes the cost of energy utilization and the cost of diverse sources of SLAV.

Xiao-Fang Liu and et.al., proposed an Ant Colony System for energy efficient Virtual machine assignment in cloud computing [9]. ACS based method is used to assign the Virtual machines in minimum numeral of physical host to diminish energy burning up for cloud data center. Also developed order exchange and migration (OEM) method to maintain together homogeneous and heterogeneous server**.**

3. Rough-Set Theory

Rough-set model is introduced by Z. Pawlak in the early 1980's, is a mathematical technique for imprecise information system and has develop into an key tool for data investigation of soft computing. Rough-set model has well-built qualitative investigation qualifications to articulate efficiently imprecise information. It has been broadly used in decision analysis, rule making, machine learning, smart control etc. The key characteristic of rough-set model is firm mathematical definitions and toughness [15]. Rough-set model can be distinct in general by the way of topological operations, and rough-set model investigation is based on inferior approximation and superior approximations are described as follows, Given a information base l=(S,R) where S is the space and R is the equivalence relation. Let x be a subset of S about information k are defined as,

$$k\_(x) = \{ v | (\forall v \in \mathrm{S}) \wedge ([v]k \subseteq v\} \quad (1)$$

$$k^{-}(x) = \{ v | (\forall v \in \mathrm{S}) \wedge ([v]k \neq \Phi\} \quad (2)$$

Where $[v]$ k specifies a correspondence class of object v about information k [16]. The inferior approximation (1) of a v with admiration to k is the set of all substance, which surely fit in to the set v on the space S based on information k. The superior approximation (2) of set v with admiration to k is the set of all substances, which possibly belong to the set v on the space S based on information k [15].

4. System Architecture

The cloud data center has unconstrained homogeneous/heterogeneous Physical machines and offers Virtual machines renting service. Figure 1 describes the system architecture, when consumer put forward their Virtual machine requests, Physical machine is identified and selected to create required Virtual machine in the data center this process is called as Virtual machine assignment. Electrical power burning up by Physical machines and supporting infrastructure in data center is considerably large. Nevertheless, requests for VMs are typically point in time variation and of some regular fashion. It is achievable to diminish power burning up by forecasting unreliable requests every so often and assigning Virtual machines consequently.

Cloud Service providers has to assure QOS as per the SLA which consequences in toggle on additional Physical machines to covenant with resource requests when required. On the other hand, turn standby some Physical machine on which all setup Virtual machine is inoperative, is a technique to diminish the power burning up. The reason behind turning standby is energy frenzied while a Physical machine is being switching on/off and not performing any useful work. The repeated switching on/off Physical machines typically requires two to three minutes as well leads to the holdup of task execution.

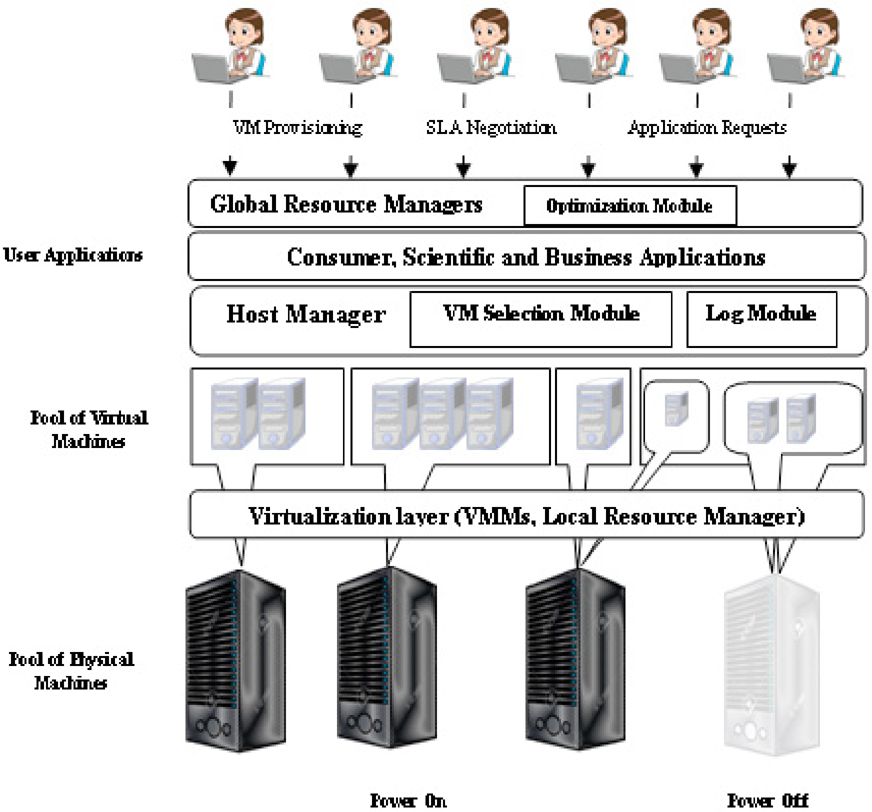

Fig. 1: System Architecture

In rough-set model Virtual machines and their attributes can be represented in a tabular form. Each tuple of the table represents list of Virtual machines and columns represents properties of the respective Virtual machine.

Table 1: sample decision information system of cloud computing

Name	ƛ1	ƛ2	ƛ3	ƛ4	ƛ5	ƛ6	ƛ7	ƛ8	δ
þм1	10	100	1024	1024	9	100	650	512	NA
þм2	10	100	1024	1024	8	98	850	626	A
þм3	10	100	1024	1024	5	95	340	626	A
þм4	10	100	1024	1024	7	65	999	512	NA
þмn	10	100	1024	1024	4	40	100	512	A

Table 1 represents the sample decision information system of cloud computing. S = { *þм1*, *þм2*, *þм3*, *þм4*, *þмn*} represents the Physical machines. v = {ƛ1, ƛ2, ƛ3, ƛ4, ƛ5, ƛ6, ƛ7, ƛ8} is the set of attributes of Physical machine, where ƛ1, ƛ2, ƛ3, ƛ4, ƛ5, ƛ6, ƛ7 and ƛ8 respectively the HostCPU, HostRam, HostDisk, HostBandwidth, UCPU, URam, UDisk, UBandwidth, D = {δ} is the decision attribute.

A decision information system expresses all the information about Physical machine model. This information table may be gratuitously large in part since it is having unnecessary or redundant values. Knowledge reduction technique reduces redundant values and also aims to uphold the categorization of the information under the convinced circumstances of reducing redundant or unnecessary information. This process of information reduction in decision information system is also known as attributes reduction [18]**.** The key idea is computing inferior and superior approximations based on Physical machines explicit characteristics and then generating the rules for consolidation of Virtual machines and it can be modelled as,

Superior approximation ($\bar{k}(x)$) = δ *(3)*

δ is the decision attribute from the reduction of all the relative attributes.

Inferior approximation ($\underline{k}(x)$) =3√ Util[k] *(4)*

Throughout Virtual machine optimization, if the exploitation of Physical machine reaches below the exploitation inferior approximation, all Virtual machines of that physical host will be migrated to the left-over switched ON Physical machine and convincing the constraint that left over Physical machine will not get over-exploited and then turn to standby the Physical machine with less exploitation. If the exploitation of Physical machine exceeds the superior approximation, then some Virtual machines of over exploited Physical machine acquire other Physical machine to migrate in order to decrease the service level agreement violation.

5. Algorithm Design

This section describes designing of Virtual machine migration algorithm that meets the QOS and SLA in addition to optimize the energy efficiency of system. Algorithm majorly determines three steps,

- **what time migration has to be happen** - Virtual machine migration can be carry out to accomplish two objectives, firstly for balancing the load, if host is overloaded then migrate Virtual machines to an under-loaded Physical machine, second to diminishes the energy frenzied by the host.
- **which Virtual machine has to migrate** - because of large numeral of Virtual machines are running on Physical machine, identification of Virtual machine to be migrate is tedious job. The Virtual machine assortment policy is used to decide the migration of particular Virtual machine.
- **where to migrate the Virtual machines selected for migration** - Virtual machine assignment policy decides the target Physical machine at which Virtual machine has to assign. The targeted Physical machine not to be supposed to overloaded past migration.

A multi-resource rough-set based technique is used to set off the Virtual machines migration.

Algorithm – Virtual machine Consolidation with Migration

Input: PMList, VMList, VMOnPMList
Output: VM is consolidated

1. PMList = GetTotalNoOfHost();{þᴍ = { þᴍ1, þᴍ2, . . . , þᴍn} ⊆ Uhs }
2. VMList = GetTotalNoOfVM(); { ʊmList = {ʊm1, ʊm2, . . . , ʊmj, . . . , ʊmm}}
3. VMOnPMList = GetCurrentNoOfVMOnHost(); { Util[k] =∑j=1,m util[j][k]}
4. Util[k]/√ ʋ (ɯ)√ ʋ (Ư) The sign ɯ and Ư refer to the current CPU exploitation of the chosen ʊm j and the host þᴍ, respectively.
5. CoHostList = null;
6. ULHostList =null;
7. for each ʊm in VMList do {
8. if (ʊmOnPMList!=null) {
9. for each Host in ʊmOnPMList do {
10. if(Host.GetUnusedCPU>=ʊm.GetRequiredCPU && Host.GetUnuesdRam >= ʊm.GetRequiredRam && Host.GetUnusedDisk >= ʊm.GetRequiredDisk)
11. { ULHostList.add(Host); }
12. if(ULHostList!=null)
13. CoHostList.add(Host);
14. ʊmOnPMList.add();

15. }}
16. Virtual machine optimization.
17. idle node -TURN TO STANDBY
18. End

6. Performance Metrics

Performance metrics is used to evaluate the algorithm efficiency. The amount of energy burning up by the Physical machines in a data center is measured as the model defined in Table 7.1. It is essential to describe the self-determining workload metric that can be used to estimate the service level agreement delivered to any Virtual machine set-up in an Infrastructure as a Service. This paper proposes two metrics for measuring the service level agreement violation level in an IaaS environment.

Violation time of SLA per Active Host (SLATH):

$$SLATH = \frac{1}{N}\sum_{i=1}^{N}\frac{T_{\partial t}}{T_{at}} \quad (5)$$

Performance dilapidation owing to Virtual machine migration

$$PDM = \frac{1}{M}\sum_{j=1}^{M}\frac{C_{dj}}{C_{rj}} \quad (6)$$

Where N is the figure shows that the numeral of physical host, $T_{\partial t}$ is the figure shows that total time of the Physical machine i which experienced over exploitation of resources leading to a service level agreement violation. T_{at} is the figure describes that total time of physical host i being in the active state, M is the figure shows that the numeral of Virtual machines, C_{dj} describes the estimation of the performance dilapidation of the Virtual machine VM_j caused by migrations. C_{rj} is the figure demonstrates that the total amount of CPU competence demanded by the Virtual machine VM_j throughout its life span. Therefore, metric designed that includes both performance dilapidation owing to physical host overloading and owing to VM migrations is anticipated and denoted as SLA Violation (SLAV) metric,

$$SLAV = SLATH * PDM \quad (7)$$

Simulations in this paper are modeled by considering the identical MIPS of CPU capacity is assigned to a Virtual machine on the target physical host throughout the course of migration and each migration causes service level agreement violation. Consequently, it is decisive to decrease the numeral of Virtual machine migrations. The total time taken for migration depends on the total amount of memory used by the Virtual machine and available network bandwidth. The Virtual machine image and data have to be stored on a NAS, and for that reason, copying the Virtual machine storage is not required. Therefore, experiments defined migration time and performance dilapidation experienced by a VM_j as

$$T_{mj} = \frac{M_j}{B_j} \qquad U_{dj} = 0.1 * \int_{t0}^{t0+Tmj} u_j(t)dt \quad (8)$$

Where U_{dj} describes that total performance dilapidation by VM_j, t_0 is the migration begin time, T_{mj} is the figure shows that time elapsed to complete the migration, $u_j(t)$ is the CPU exploitation by VM_j, M_j is the figure describes that the capacity of memory used by VM_j, and B_j is the accessible network bandwidth.

The estimation of migration time is as the capacity of memory (RAM) used by VM divided by the auxiliary network bandwidth accessible for the host j. Let Vj be a set of VMs currently assigned to the host j.

$$v \in V_j \mid \forall \in V_j, \frac{R_u(v)}{N_j} \leq \frac{R_u(a)}{N_j} \quad (9)$$

where R_u (a) is the capacity of memory (RAM) presently used by the VM a, and N_j is the auxiliary network bandwidth accessible for host j.

7. Experiments and Results

It is exceptionally intricate to carry out reiterated bulky scale experimentation on a real infrastructure. The CloudSim tool-kit [14] has been preferred platform for simulation. CloudSim tool-kit is designed and implemented by Melbourne University and is an extensible simulator, whose objective is to facilitate the simulation and modeling environment for cloud computing. It allows simulating virtualized resources, cloud computing entities like data centers, Virtual machines, and physical hosts. Thus CloudSim tool-kit facilitates to implement diverse resource assignment policies and estimate the policy performance.

This work carried out a simulation of data center, which consists of 800 homogeneous and heterogeneous Physical machines, half of which are HP Pro ML 110 G4 servers, and remaining are HP Pro ML 110 G5 servers. Power consumption of servers is described in Table 2. The server CPU frequency is mapped onto MIPS ratings, HP Pro ML 110 G4 server has 1860 MIPS each core and HP Pro ML 110 G5 server has 2660 MIPS each core and each server network bandwidth is modeled to 1 GBPS.

Table 2: Power frenzied by the servers at diverse load levels in Watts

Server/Power Consumption	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
HP Pro ML G4	86	89.4	92.6	96	99.5	102	106	108	112	114	117
HP Pro ML G5	93.7	97	101	105	110	116	121	125	129	133	135

Conduction of experiments, used data offered by monitoring infrastructure for PlanetLab[13]. Throughout the experimentation, each Virtual machine is arbitrarily allotted a workload outline from one of the Virtual machines from the consequent day and interval time of monitoring is 5 minutes. System Performance Metrics are described in Table 3, proposed system is simulated for 1hr by considering 800 hosts and 1033 Virtual machines. Figure 2 – 5 shows inference and results drawn for the proposed system, also system performance is compared with the different approaches available.

Table. 3: System Performance Metrics

	Performance Metrics			**Performance Metrics**	
1.	Energy Consumed	2.4	6.	Overall Violation	0.89
2.	No. of Migrations	705	7.	Avg. SLA Violation	10.02
3.	SLA	0.064	8.	No. of Stand By	293
4.	SLA Degradation	0.75	9.	Mean time before shutdown	349.76
5.	SLATH	8.57	10.	St.Dev time before host shutdown	84.86

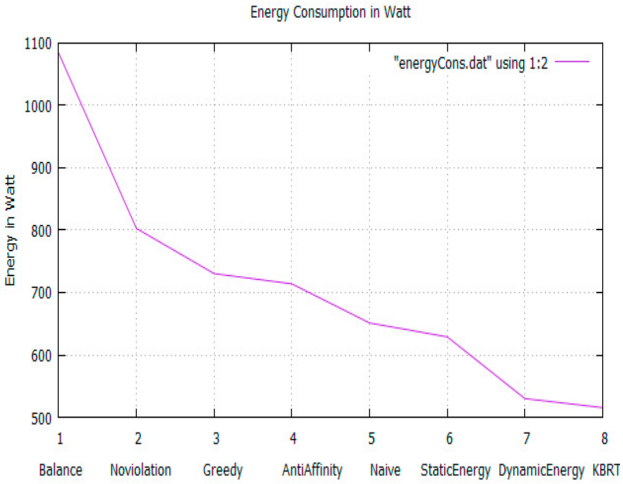

Fig. 2: shows the comparision of enrgy consumption in kwh

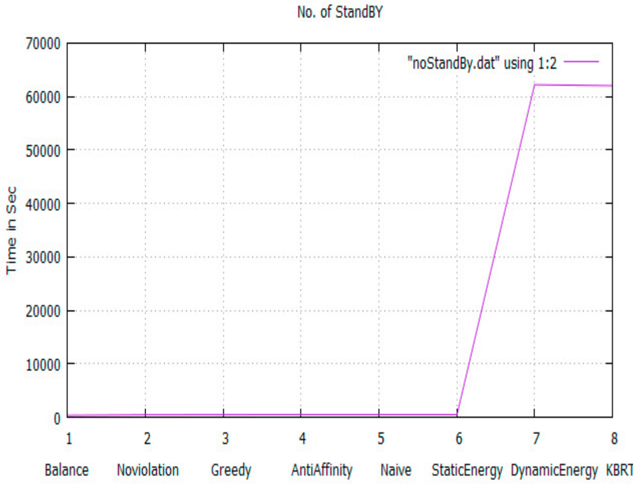

Fig. 3: shows numeral of Physical machine turns to standby mode and mean time before standby

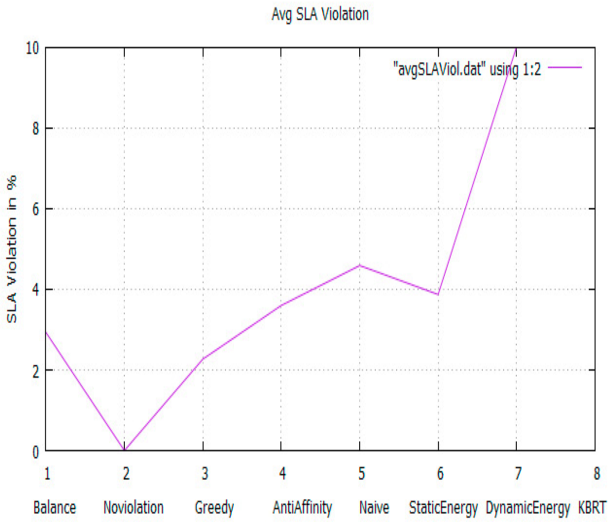

Fig. 4: shows the comparision of SLA violation

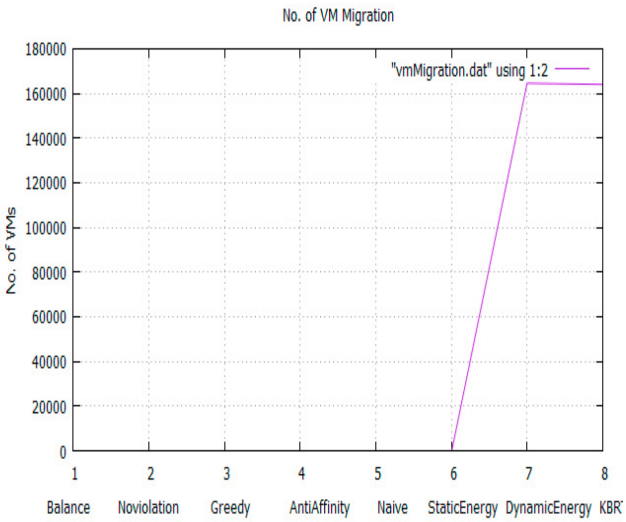

Fig. 5: shows numeral of Virtual machine migration

Conclusion

This paper addresses the problem of Virtual machine assignment and consolidation with the objective of diminishing the total energy burning up by Physical machine. This work proposed knowledge based reduction method for Virtual machine assignment and consolidation. Proposed method computes superior and inferior approximation for exploiting resource exploitation, which helps in reducing numeral of Virtual machine migration happen in cloud computing environment, also lower down the numeral of running Physical machines and therefore diminishes the energy burning up. The algorithm usefulness and competence is validated by conducting experiments.

References

[1] Multi-Objective VM Consolidation Based on Thresholds and Ant Colony System in Cloud Computing, Hui Xiao, Zhigang Hu and Keqin Li, IEEE Transactions, April 2019, DOI:10.1109/Access.2019.9.2912722.

[2] An experience-Based Scheme for Energy-SLA Balance in Cloud Data Centers, Xijia Zhou, Kenli Li, Chubo Liu, Keqin Li, IEEE Transactions, March-2019, DOI: 10.1109/Access.2019.2899101.

[3] Energy-Aware Dynamic Virtual Machine Consolidation for Cloud Data Centers, Hui Wang and Huaglory Tianfield, IEEE Transactions, March-2018, DOI:10.1109/Access.2018.2813541.

[4] Energy-Efficient Virtual Resource Dynamic Integration Method in Cloud Computing, Yingyou Wen, Zhi Li, Shuyuan jin, Chuan Lin and Zheng Liu, IEEE Transactions Special Section on Emerging Trends, Issues, and Challenges and Energy-Efficient Cloud Computing, June-2017, DOI:10.1109/Access.2017.2721548.

[5] Energy-aware Virtual Machine Migration for Cloud Computing- A Firefly Optimization Approach, Journal of Grid Computing, Nidhi Jain Kansal and Inderveer Chana, June 2016, Volume 14, Issue 2, pp 327 – 345.

[6] An Efficient Virtual Machine Migration Algorithm based on Minimization of Migration in Cloud Computing, Nguyen Khac Chien, Vo Sy Giang Dong, Nguyen Hong Son, Ho Dac Loc, International Conference on Nature of Computation and Communication, Springer, March 2016, DOI:10.1007/978-3-319-46090-6_7.

[7] Layered virtal machine migration for network resource balancing in cloud computing, Xiong Fu, Juzhon Chen, Song Deng, Junchang Wang, Lin Zhang, Frontiers of Computer Science, February 2018, Volume 12, Issue 1, pp 75–85.

[8] Optimizing virtual machine placement for energy and SLA in clouds using utility functions, Journal of cloud computing: Advances, Systems and Applications 2016, DOI 10.1186/s13677-016-0067-7.

[9] An Energy Efficient Ant Colony System for Virtual Machine Placement in Cloud Compiting, Xiao-Fang Liu, Yun Li, Tianlong Gu, Jun Zhang, IEEE Transaction on Evolutionary Computation, vol-22, Feb-2018, pp 113-128.

[10] Virtual machine selection and placement for dynamic consolidation in Cloud computing environment, Xiong Fu, Frontiers of Computer Science, April 2015, Volume 9, Issue 2, pp 322–330.

[11] Energy efficient virtual machine placement algorithm with balanced and improved resource exploitation in a data center, Xin Li a, Zhuzhong Qian, Sanglu Lua, Jie Wub, Mathematical and Computer Modelling 58 (2013) 1222–1235, doi:10.1016/j.mcm.2013.02.003

[12] Energy Efficient Allocation of Virtual Machines in Cloud Computing Environments Based on Demand Forecast, Jian Cao, Yihua Wu, and Minglu Li, Springer-Verlag Berlin Heidelberg 2012, pp. 137–151.

[13] CoMon: a mostly-scalable monitoring system for PlanetLab, Park KS, Pai VS, ACM SIGOPS Operating Systems Review 2006; 40(1):74

[14] CloudSim: a toolkit for modeling and simulation of Cloud computing environments and evaluation of resource provisioning algorithms, Calheiros RN, Ranjan R, Beloglazov A, Rose CAFD, Buyya R, Software: Practice and Experience 2011; 41(1):23–50.

[15] Rough set theory and its applications , Zdzisław Pawlak , Journal of Telecommunication and Information Technology, March 2012.

[16] Cloud Service Selection Based on Rough Set Theory, Yongwen Liu, Moez Esseghir, Leila Merghem Boulahia, IEEE.

[17] Rough Set Theory-fundamental concepts , principles, data extraction and applications, S. Rissino and G. Lambort-Torres, Data mining and knowledge discovery in real lite applications, pp.293-299, 2010.

[18] Knowledge Based Reduction Technique for Virtual Machine Provisioning in Cloud Computing, Bhaskar R, Dr. Shylaja B S, International Journal of computer science and information security(IJCSIS), vol 24, No 7, ISSN 1947-5500 , July 2016.

[19] Dynamic Allocation Method for Efficient Load Balancing in Virtual Machines for Cloud Computing, Bhaskar. R, Deepu. S.R, Dr. Shylaja B.S, Advanced Computing: An International Journal(ACIJ), Vol.3, No.5, September 2012.